Affine groups after one and two halvings

Human Compression project

The translation lattices of cubic toroids give a direct way to compute their halving groups. One first takes the subgroup generated by the new reflections in the infinite affine group, and then intersects its translation lattice with the original period lattice. This gives all the groups in Propositions 5.8–5.13 of the inspected preprint [1] by one calculation. It also detects missing relations in several of the claimed presentations. We distinguish the calculation of a marked group from the geometric conditions required to perform a halving.

1 Affine notation and the two subgroups

Let $n \geq 3$. Let B_n be the signed permutation group on $\mathbb{Z}^n$, and let B_n^+ be its subgroup with an even number of negative signs. Thus $|B_n| = 2^n n!$ and $|B_n^+| = 2^{n-1} n!$. Let s_i exchange coordinates $i, i+1$, let z_i change the sign of coordinate i, and let τ_v denote translation by v. We multiply affine transformations by composition, with the rightmost factor acting first. The usual cubic marking of $W = \mathbb{Z}^n \rtimes B_n$ is

$$a = \tau_{e_1} z_1, \qquad s_1, \ldots, s_{n-1}, \qquad z_n.$$

We retain its standard affine Coxeter presentation $[4, 3^{n-2}, 4]$. This is the same classical realization used in Theorem 5.1 of [1]. Put

$$D = \{v \in \mathbb{Z}^n : \textstyle\sum_i v_i \equiv 0 \pmod 2\}, \quad x = a s_1 a = \tau_{e_1+e_2} z_1 z_2 s_1, \quad y = z_n s_{n-1} z_n = z_{n-1} z_n s_{n-1}.$$

The new markings after one and two endpoint operations generate

$$K = \langle x, s_1, \ldots, s_{n-1}, z_n \rangle, \qquad J = \langle x, s_1, \ldots, s_{n-1}, y \rangle.$$

Lemma 1. *One has $K = D \rtimes B_n$ and $J = D \rtimes B_n^+$. The Coxeter presentations of these groups are, respectively, the diagram obtained by halving the first endpoint of $[4, 3^{n-2}, 4]$, and the diagram obtained by halving both endpoints. In rank four the latter is a square of bonds labelled three; in rank five it is the four-leaf diagram with all bonds labelled three.*

Proof. The linear generators of K generate B_n. Multiplying x by $z_1 z_2 s_1$ gives $\tau_{e_1+e_2}$. Its conjugates by B_n are translations by all vectors $\pm e_i \pm e_j$, which span D. Conversely every displayed generator belongs to $D \rtimes B_n$. For J, the generators s_i and y generate B_n^+: the adjacent swaps give all permutations, while $y s_{n-1}$ changes two signs and its conjugates give every pair of sign changes. Again $z_1 z_2 s_1 \in B_n^+$, so $\tau_{e_1+e_2} \in J$. The orbit of this vector under B_n^+ contains all $\pm e_i \pm e_j$ when $n \geq 3$. To change just one of its two signs one changes the sign of a third, zero coordinate as well. This proves the second equality.

Here is also a presentation argument, needed when comparing the printed extra relations. In a Coxeter presentation with an endpoint a joined to b by a bond four and commuting with the remaining generators, replace a by $x = aba$. The proposed new Coxeter diagram duplicates the bonds of b at x, and makes x, b commute. Its abstract Coxeter group has an involutory automorphism exchanging x, b and fixing the other generators. Adjoin an involution a acting by this automorphism. Eliminate x using $x = aba$. The relation $(xb)^2 = 1$ becomes $(ab)^4 = 1$; the remaining relations at x are conjugates of those at b. All other relations are exactly the old Coxeter relations. Thus the old group is the semidirect product of the new Coxeter group by $\langle a \rangle$, and the new group embeds as the kernel of the endpoint parity character. Apply this argument first at a in W, then at z_n in K. In the second application the character is the parity of the number of negative signs. This gives precisely the groups already identified and also covers $n = 3, 4$. □

2 The period lattice is intersected, not merely squared

For $s \geq 2$, the three toroid period lattices are

$$\Lambda_1 = s\mathbb{Z}^n, \qquad \Lambda_2 = sD, \qquad \Lambda_n = s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}), \qquad \mathbf{1} = (1, \ldots, 1).$$

These are the spans of the signed permutation orbits of se_1, $s(e_1 + e_2)$ and $s\mathbf{1}$, respectively, as in [1]. They are normal translation subgroups of W. Let $\overline{K}$ and $\overline{J}$ denote the images of the two marked groups in W/Λ_k.

Theorem 2. *Set $L = D \cap \Lambda_k$. Then, with their displayed affine markings,*

$$\overline{K} = (D/L) \rtimes B_n, \qquad \overline{J} = (D/L) \rtimes B_n^+.$$

The exact lattice and its index in D are as follows.

parameters	L	$[D : L]$
$k = 1,\ s\ even$	$s\mathbb{Z}^n$	$s^n/2$
$k = 1,\ s\ odd$	sD	s^n
$k = 2$	sD	s^n
$k = n,\ ns\ even$	$s(2\mathbb{Z}^n + \mathbb{Z}\mathbf{1})$	$2^{n-2}s^n$
$k = n,\ ns\ odd$	$2s\mathbb{Z}^n$	$2^{n-1}s^n$

In particular $|\overline{K}| = 2^n n![D : L]$ and $|\overline{J}| = 2^{n-1}n![D : L]$.

Proof. Restriction of the quotient homomorphism $W \to W/\Lambda_k$ to either group in Lemma 1 has kernel $D \cap \Lambda_k$. The linear subgroup intersects this kernel trivially. It therefore remains a complement, proving the two semidirect product statements.

A vector sv has even coordinate sum automatically when s is even, and exactly when $v \in D$ when s is odd. This gives the first two rows. The third is immediate. Write an element of Λ_n as $s(2v + t\mathbf{1})$. Its coordinate sum modulo two is nst. If ns is even there is no restriction; otherwise t must be even, giving $2s\mathbb{Z}^n$. Finally $[\mathbb{Z}^n : D] = 2$ and $[\mathbb{Z}^n : 2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}] = 2^{n-1}$. Multiplication of a full lattice by s multiplies its index by s^n. These facts give all the displayed indices. □

This is a complete group description, rather than only an order calculation. It also supplies finite presentations uniformly. Choose the basis of D

$$d_1 = e_1 + e_2, \quad d_2 = e_2 - e_1, \quad d_i = e_i - e_{i-1} \quad (3 \leq i \leq n).$$

Adjoin commuting translation generators $T_1, \ldots, T_n$ to a standard presentation of the finite group $P = B_n$ or B_n^+. For each generator p of P impose

$$pT_j p^{-1} = \prod_{i=1}^{n} T_i^{c_{ij}(p)}, \qquad pd_j = \sum_i c_{ij}(p) d_i.$$

For a basis of L impose $\prod_i T_i^{\ell_i} = 1$ whenever $\sum_i \ell_i d_i$ is a member of that basis. Such a basis is se_i in the first row, sd_i in the second and third, $2se_1, \ldots, 2se_{n-1}, s\mathbf{1}$ in the fourth, and $2se_i$ in the fifth. All these vectors lie in D under the stated parity assumptions, so their coefficients are integers. The conjugation relations move every word to the form $T^v p$. The multiplication law and the evident affine realization then show that these presentations give exactly $(D/L) \rtimes P$. The marked element x is $T_1 z_1 z_2 s_1$, and the other marked elements are the specified linear transformations. Thus the presentation retains the actual marking as well as the abstract group.

3 Comparison with the single extra relations

The defect in a proposed extra relation can now be tested inside the infinite groups, before taking any finite quotient. A pure translation τ_v normally generates the translations whose vectors lie in the integer span of the orbit of v under the linear complement. Translations conjugate one another trivially, so there are no additional conjugates to supply missing periods.

For clarity we evaluate every family of extra words used in Propositions 5.8–5.13. Put

$$u = as_1 \cdots s_{n-1}z_n, \qquad u(X_1, \ldots, X_n) = (X_n + 1, X_1, \ldots, X_{n-1}).$$

Thus $u^n = \tau_{\mathbf{1}}$. The words for $k = n$, both before and after the second halving, are u^2 written in the new generators. This follows by moving a past s_i for $i \geq 2$ in the first case; in the second, move z_n past s_i for $i < n - 1$ and replace $z_n s_{n-1} z_n$ by y. The resulting word is

$$xs_2 \cdots s_{n-1}s_1s_2 \cdots s_{n-2}y.$$

Its power $ns/2$, when integral, is $\tau_{s\mathbf{1}}$; its power ns in the odd case is $\tau_{2s\mathbf{1}}$.

For $k = 1$, the single-halving word is

$$q = xs_2 \cdots s_{n-1}z_n s_{n-1} \cdots s_2 s_1 = \tau_{e_1+e_2} z_2.$$

The corresponding word after two halvings is

$$q' = xs_2 \cdots s_{n-1}ys_{n-2} \cdots s_2 s_1 = \tau_{e_1+e_2} z_2 z_n.$$

Both square to τ_{2e_1}. Thus their displayed powers are τ_{se_1} for even s and τ_{2se_1} for odd s.

For $k = 2$, put

$$A = s_2 \cdots s_{n-1}z_n s_{n-1} \cdots s_2 = z_2, \qquad A' = s_2 \cdots s_{n-1}ys_{n-2} \cdots s_2 = z_2 z_n.$$

These concern the first and second operation, respectively. The words xAs_1A and $xA's_1A'$ are both $\tau_{e_1+e_2}$. The rank-four word printed separately in Proposition 5.12 is $xs_2s_1ys_1s_2$; conjugating y by s_2s_1 shows its linear part is $z_1z_2s_1$, so it has the same value. Consequently the extra relator is the translation $\tau_{s(e_1+e_2)}$ in every $k = 2$ case.

Proposition 3. *Interpret the listed extra relations in [1] as complete presentations, as their proofs do. They have the following exact discrepancies. The factor is the order of the presented group divided by the order of the intended marked group.*

parameters	*one halving*	*two halvings*
$k = 1$, *s odd*	2^{n-1}	2^{n-1}
$k = n$, *ns odd*	2^{n-1}	2^{n-1}
$k = n$, *n even*	1	2
all other cases	1	1

The table compares presentations of marked groups. Geometric validity of the second operation is a separate hypothesis.

Proof. The signed permutation orbit of e_1 spans $\mathbb{Z}^n$ under either complement, and that of $e_1 + e_2$ spans D. The orbit of $\mathbf{1}$ under B_n spans $E = 2\mathbb{Z}^n + \mathbb{Z}\mathbf{1}$: changing one sign gives $2e_i$, and these vectors together with $\mathbf{1}$ also generate every orbit vector. Its orbit under B_n^+ instead spans

$$M = 2D + \mathbb{Z}\mathbf{1}.$$

Indeed differences from $\mathbf{1}$ are twice the characteristic vectors of even subsets; the two-element subsets already span D. If n is odd, then $\mathbf{1} - e_i \in D$, so $2e_i = 2\mathbf{1} - 2(\mathbf{1} - e_i) \in M$ and $M = E$. If n is even, M has index two in E. To see this, every generator $2e_i$ of $2\mathbb{Z}^n$ has the same coset modulo $2D$, so the

index is at most two. It is not one: an equality $2e_1 = 2d + t\mathbf{1}$, with $d \in D$, forces $t = 2h$ and then $e_1 = d + h\mathbf{1}$, contradicting parity of coordinate sums.

Let R be the normal translation lattice generated by the printed extra relation. For odd s and $k = 1$, one has $R = 2s\mathbb{Z}^n$, whereas $L = sD$. Their relative index is $[sD : 2s\mathbb{Z}^n] = 2^{n-1}$. For odd ns and $k = n$, one has $R = 2sE$ for either operation, whereas $L = 2s\mathbb{Z}^n$; again the relative index is 2^{n-1}. For even n and $k = n$, the first operation gives $R = sE = L$, while the second gives $R = sM$, of index two in $L = sE$. The other rows give $R = L$ by the same orbit calculations. Lemma 1 identifies the underlying Coxeter group exactly, so its quotient by the displayed relation is $(D/R) \rtimes P$. The factor is therefore exactly $[L : R]$, not merely a lower bound from a finite quotient. □

For example, at $n = 3, k = 1, s = 3$, the intended first halving group has order $3^3 \cdot 48 = 1296$. The displayed presentation instead has order $[D : 6\mathbb{Z}^3] \cdot 48 = 108 \cdot 48 = 5184$. The accompanying small diagnostic constructs both affine groups directly, verifies every Coxeter relation and the displayed extra word, and obtains these orders. The all-parameter conclusions above follow from the lattice proofs, not from this finite check.

4 Attribution and remaining geometric obligations

The cubic tessellation, the three period lattices and the halving operations are prior work. The parity characters, the second isomorphism theorem, and the standard Coxeter presentations are also established ingredients. Our contribution here is their common application to the exact period intersection, the complete presentations that it supplies, and the comparison that identifies the missing relations in the inspected edition. We make no claim of publication priority or about a later journal revision. The assertions that particular iterated constructions remain hypertopes, and the full list of their diagrams, require the separate nondegeneracy and residue arguments. This note does not count N014 as a completed recovery.

References

[1] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1 (29 May 2024), Section 5, especially Propositions 5.8–5.13.